

Statistical Analysis Plan

Title: Water Source and Diarrhoeal Disease: A Prospective Cohort Study in a Simulated District Population
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1. Study Objectives

1.1 Primary Objective

To estimate and compare the incidence of diarrhoea among individuals using well water versus tap water as their primary household water source.

1.2 Secondary Objectives

1. To describe the baseline sociodemographic and household characteristics of the study cohort, stratified by water source.
2. To estimate the cumulative incidence (risk) of diarrhoea (at least one episode during follow-up).
3. To estimate the overall incidence rate of diarrhoea (accounting for recurrent episodes and total person-time).
4. To estimate crude and adjusted measures of association (Risk Ratios, Rate Ratios, Hazard Ratios, and Odds Ratios) comparing exposed (well water) and unexposed (tap water) groups.
5. To perform time-to-first-episode survival analysis.

2. Variables and Data Definitions

The analysis will use the dataset `wsdd_data.csv`. The variables are defined as follows:

Variable Name	Description	Data Type	Operationalization & Coding
id	Participant ID	Integer	Sequential identifier (1–1,200)
enrolment_date	Date of enrolment	Date	Date format (YYYY-MM-DD); used to assess seasonality or secular trends
age	Age at enrolment	Integer	Continuous (1–85 years); will be

Variable Name	Description	Data Type	Operationalization & Coding
			modeled continuously and categorized if non-linear
sex	Biological sex	Factor	Binary: 0 = Male (ref), 1 = Female
water_source	Primary water source (Exposure)	Factor	Binary: 0 = Tap water (ref), 1 = Well water
sanitation	Improved sanitation access	Factor	Binary: 0 = No (ref), 1 = Yes
household_size	Number of household members	Integer	Continuous (1–15)
socioeconomic_status	Household socioeconomic status	Factor	Ordinal: 1 = Low (ref), 2 = Medium, 3 = High
education_head	Education of household head	Factor	Ordinal: 1 = None (ref), 2 = Primary, 3 = Secondary, 4 = Tertiary
num_episodes	Total number of diarrhoea episodes	Integer	Count (0, 1, 2, ...); used for incidence rates and recurrent event modeling
time_to_first_episode	Time to first diarrhoea episode	Integer	Continuous (1–365 days); censored at follow_up_time if no episode
follow_up_time	Total follow-up time	Integer	Continuous (1–365 days)
lost_to_follow_up	Lost to follow-up status	Factor	Binary: 0 = Completed (ref), 1 = Lost

2.1 Derived Variables

- **Diarrhoea Status (diarrhoea_status):** A binary variable indicating whether a participant experienced at least one episode of diarrhoea during the follow-up period.

$$\text{diarrhoea_status} = \begin{cases} 1 & \text{if num_episodes} > 0 \\ 0 & \text{if num_episodes} = 0 \end{cases}$$

- **Net Person-Time at Risk (Days):** The follow-up time adjusted by subtracting the time the participant was actively ill. Following clinical assumptions, each diarrhoeal episode is assumed to last an average of 3 days.

$$\text{net_person_time_days} = \text{follow_up_time} - (\text{num_episodes} \times 3)$$

- **Person-Time at Risk (Years):**

$$\text{person_time_years} = \frac{\text{net_person_time_days}}{365.25}$$

3. Data Cleaning and Preprocessing

Prior to statistical analysis, data cleaning will be performed in `s01_clean_data.r`. The following checks and actions will be implemented:

- **Logic Inconsistencies:**
 - Check if `num_episodes > 0` but `time_to_first_episode` is NA (or vice versa).
 - Check if `time_to_first_episode > follow_up_time`.
 - Check if `lost_to_follow_up = 0` but `follow_up_time < 365`.
 - Check for negative values in age, household size, or follow-up time.
- **Factor Re-leveling:**
 - All categorical variables will be explicitly converted to factors. Reference levels will be set using `{forcats}` to represent the unexposed or lowest risk group (e.g., Tap water for exposure, Male for sex, Low for SES).

4. Statistical Methods

4.1 Descriptive Analysis

Baseline demographics and household characteristics will be described and stratified by `water_source`.

- **Continuous Variables:** Summarized as mean (Standard Deviation, SD) for normally distributed variables, or median (Interquartile Range, IQR) for skewed variables. Stratified comparisons will use the independent-samples t-test or Wilcoxon rank-sum test.
- **Categorical Variables:** Summarized as frequencies (*n*) and percentages (%). Comparisons will use the Chi-squared test or Fisher's exact test (if cell counts are < 5).
- **Reporting:** A baseline Table 1 will be generated using `gtsummary::tbl_summary` with an overall column and explicit reporting of missing data (`missing = "ifany"`).

4.2 Primary Analysis: Crude Estimates

- **Cumulative Incidence (Risk):**

- Cumulative incidence proportion reported overall and stratified by water source with exact binomial (Clopper-Pearson) 95% Confidence Intervals (CIs).
- **Risk Ratio (RR):** Calculated as the risk in the well water group divided by the risk in the tap water group, with 95% Wald CI.
- **Incidence Rate (IR):**
 - Incidence rate computed overall and stratified by water source as the total number of episodes divided by total person-time at risk in years, expressed per 1,000 person-years, with Poisson 95% CIs.
 - **Incidence Rate Ratio (IRR):** Calculated as the rate in the well water group divided by the rate in the tap water group, with Poisson 95% CI.

4.3 Multivariable Modeling & Confounder Adjustment

To control for confounding, covariates will be selected a priori based on clinical knowledge and epidemiological plausibility. No automated stepwise selection methods will be used. Confounders adjusted for will include: age, sex, sanitation, socioeconomic_status, and education_head.

- **Risk of Diarrhoea (Binary Outcome):**
 - **Model:** Multivariable Logistic Regression.
 - **Estimates:** Adjusted Odds Ratios (aOR) with 95% profile-likelihood CIs.
 - **Alternative (for Direct Risk Ratio Estimation):** Multivariable log-binomial regression or Poisson regression with robust standard errors (sandwich estimator) to compute Adjusted Risk Ratios (aRR).
- **Rate of Diarrhoea (Count Outcome):**
 - **Model:** Negative Binomial Regression (preferred over standard Poisson if overdispersion is present, which will be evaluated using the dispersion parameter test).
 - **Offset:** $\log(\text{person_time_years})$ will be included as an offset term.
 - **Robust Standard Errors:** Since participants can experience recurrent events, we will calculate robust standard errors to account for within-person correlation.
 - **Estimates:** Adjusted Incidence Rate Ratios (aIRR) with 95% CIs.

4.4 Time-to-Event (Survival) Analysis

To model the time from enrolment to the first diarrhoea episode:

- **Kaplan-Meier Estimator:** Survival curves showing the probability of remaining free of diarrhoea over time, stratified by water_source. Statistical differences will be evaluated using the log-rank test.
- **Cox Proportional Hazards Regression:**
 - Proportional Hazards (PH) models will estimate crude and adjusted Hazard Ratios (cHR, aHR) and 95% CIs.

- **PH Assumption Verification:** Tested using Schoenfeld residuals via `survival::cox.zph()`. If the assumption is violated for a covariate, stratified Cox models or time-dependent covariates will be introduced.

4.5 Diagnostics and Collinearity

- Multicollinearity among predictors will be evaluated using the Variance Inflation Factor (VIF) from the logistic and survival models. A $VIF > 5$ will warrant further investigation.

5. Handling of Missing Data

- **Evaluation:** Missing data patterns will be visualized using `naniar::vis_miss()`.
- **Strategy:**
 - Complete-case analysis will serve as the primary approach.
 - If missingness in key covariates (e.g., `education_head`, `socioeconomic_status`) exceeds 5%, Multiple Imputation by Chained Equations (`mice::mice()`) will be conducted as a sensitivity analysis. Ten imputed datasets will be generated, and model results pooled using Rubin's rules.

6. Reporting and Presentation Standards

- **P-values:** Reported to 3 decimal places (e.g., $p = 0.024$). P-values less than 0.001 will be reported as $p < 0.001$.
- **Effect Sizes:** All regression results will be accompanied by relative effect sizes (aOR, aRR, aIRR, aHR) and their corresponding 95% CIs.
- **Visualizations:** Figures will be created using `ggplot2` using color-blind friendly palettes (e.g., `scale_color_viridis_d()`) and `theme_minimal()`. Kaplan-Meier plots will include risk tables using `survminer::ggsurvplot`.
- **Reproducibility:** Code syntax will be formatted via `styler::style_dir()` and checked using `lintr::lint_dir()`. The R session information will be appended to all final reports to track package versions.